Curriculum Reinforcement Learning for Variable-Height Balancing of a 10-Actuated Wheeled Biped Robot

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***Abstract*—This paper presents a curriculum reinforcement learning framework for training a 10-actuated wheeled biped robot to balance at variable commanded torso heights. Each leg has five actuated joints: hip roll, hip yaw, hip pitch, knee, and wheel. The control policy is trained in MuJoCo MJX with JAX-accelerated vectorized simulation using proximal policy optimization (PPO). The policy receives a 41-dimensional observation that includes proprioceptive state, a normalized height command, and a yaw-error signal, and outputs joint-level targets that are converted to actuator torques through a PID low-level controller. To stabilize learning across a wide range of body heights, we employ a within-stage height curriculum that progressively expands the commanded height range from near the nominal standing posture toward deep crouching configurations, gated by a reward-threshold criterion evaluated on a dedicated greedy evaluation pass every $I_{\text{eval}} = 2$ training updates. A key component of the reward design is a height-consistent natural pose term that maps commanded height to target hip-pitch and knee angles via piecewise-linear interpolation, preventing degenerate postures that satisfy height objectives through unnatural joint configurations. We additionally implement a classical LQR baseline built on a linearized two-wheeled inverted pendulum (TWIP) model with height-dependent IK, providing a fair reference for the near-nominal operating range. The framework also includes a multi-stage macro-curriculum structure (stages: `balance`, `stand_up`, `balance_robust`) for future extension, though the present work focuses on the variable-height balancing stage. We describe the full system design, reward formulation, curriculum strategy, LQR baseline, and evaluation protocol, and present the framework as a foundation for further training and experimental validation.**

***Index Terms*—Wheeled biped robot, reinforcement learning, curriculum learning, variable-height balancing, PID control, LQR baseline, MuJoCo MJX.**

I. Introduction

Wheeled biped robots combine the efficiency of wheel-based locomotion with the adaptability of legged morphology. Compared with conventional legged systems, wheeled bipeds can achieve lower-cost motion on flat terrain while exploiting articulated legs to modify body posture, regulate center-of-mass height, and improve disturbance rejection. However, these benefits come at the cost of more challenging control: balancing at different commanded heights requires managing the nonlinear coupling between wheel dynamics, torso attitude, and multi-joint leg motions.

Classical control approaches [11] can stabilize wheeled bipeds near a nominal operating point, but become increasingly inaccurate as the robot moves away from the linearization height. Recent progress in reinforcement learning (RL) has shown that robust control policies can be learned directly in simulation for complex robotic systems [1]. Nevertheless, training a single policy to balance across a wide range of body heights is difficult due to poor sample efficiency and the risk of reward exploitation. In particular, simple height-tracking rewards may allow policies to satisfy torso height objectives through unnatural or unstable joint configurations rather than through physically meaningful postures.

To address these challenges, we propose a curriculum reinforcement learning framework for a 10-actuated wheeled biped robot. The method explicitly conditions the policy on a target torso height and uses a within-stage height curriculum to progressively expand the difficulty from near-nominal standing to deep crouching. A height-consistent posture reward regularizes the joint configuration as a function of the commanded height, improving training stability and posture quality. We also implement a classical LQR baseline that operates on the same observation interface and PID low-level controller as the RL policy, providing an interpretable reference for the regime where linear assumptions hold.

The main contributions of this work are:

- A height-conditioned RL framework for variable-height balancing of a 10-actuated wheeled biped, with PID low-level control bridging policy outputs to actuator torques.
- A within-stage height curriculum that expands the commanded height range based on a greedy evaluation pass gated at a reward-threshold criterion, stabilizing learning across the full feasible height range.
- A height-consistent natural pose reward that ties hip-pitch and knee joint targets to the commanded torso height via piecewise-linear interpolation, preventing degenerate balancing strategies.
- A classical LQR baseline built on a linearized TWIP model with height-dependent inverse kinematics, serving as an interpretable comparison for the near-nominal height regime.
- A quantitative evaluation benchmark with ten scenario groups and ten per-scenario metrics, enabling rigorous comparison between the RL policy and the LQR baseline.

The framework additionally includes a multi-stage

macro-curriculum (`balance` → `stand_up` → `balance_robust`) and a research evaluation benchmark suite implemented in `scripts/eval_balance.py`. These extensions are described in this paper but their full experimental results remain as future work.

The rest of this paper is organized as follows. Section II reviews related work. Section III presents the robot platform and simulation setup. Section IV formulates the learning problem. Section V describes the reward design. Section VI presents the curriculum strategy. Section VII describes the classical LQR baseline. Section VIII details the training pipeline and evaluation protocol. Section IX is reserved for experimental results. Section X discusses limitations and future work. Section XI concludes the paper.

II. Related Work

Wheeled bipedal robots have been studied in several recent works. Klemm *et al.* [10] developed Ascento, a two-wheeled jumping robot, demonstrating the feasibility of active balance control for wheeled bipeds. In subsequent work, Klemm *et al.* [11] proposed an LQR-assisted whole-body control approach for a wheeled bipedal robot with kinematic loops, achieving stable balancing through model-based control. Hsu *et al.* [12] applied fuzzy logic control to a wheeled biped for balance regulation. Li *et al.* [13] investigated dynamic wheeled motion control for wheel-biped transformable robots.

On the learning side, deep reinforcement learning has enabled robust locomotion for legged robots [1]. Curriculum learning approaches [9] have been applied to progressively increase task difficulty, improving sample efficiency and the range of behaviors learned.

These approaches predominantly rely on model-based control or classical control techniques. In contrast, our work takes a learning-based approach using PPO [1] in a GPU-accelerated MuJoCo MJX [2], [3] simulation with JAX [4], which enables massively parallel training across thousands of environments. The key distinction of our framework is the combination of a height-conditioned policy, a progressive height curriculum, and a height-consistent posture reward that together address the challenge of stable balancing across a wide range of body heights, alongside an LQR baseline that provides a principled classical reference at the nominal operating point.

III. Robot Platform and Simulation Setup

A. Robot Morphology

The robot considered in this work is a 10-actuated wheeled biped with five actuated joints on each leg: hip roll, hip yaw, hip pitch, knee, and wheel. The platform has a total mass of approximately $8.1\,\mathrm{kg}$, thigh length of $0.26\,\mathrm{m}$, shin length of $0.28\,\mathrm{m}$, wheel radius of $0.06\,\mathrm{m}$, and hip width of $0.23\,\mathrm{m}$. The sensing system includes one IMU and ten joint encoders. A physical robot with the same kinematic structure exists and is intended for future sim-to-real transfer, though hardware experiments are not part of the present work.

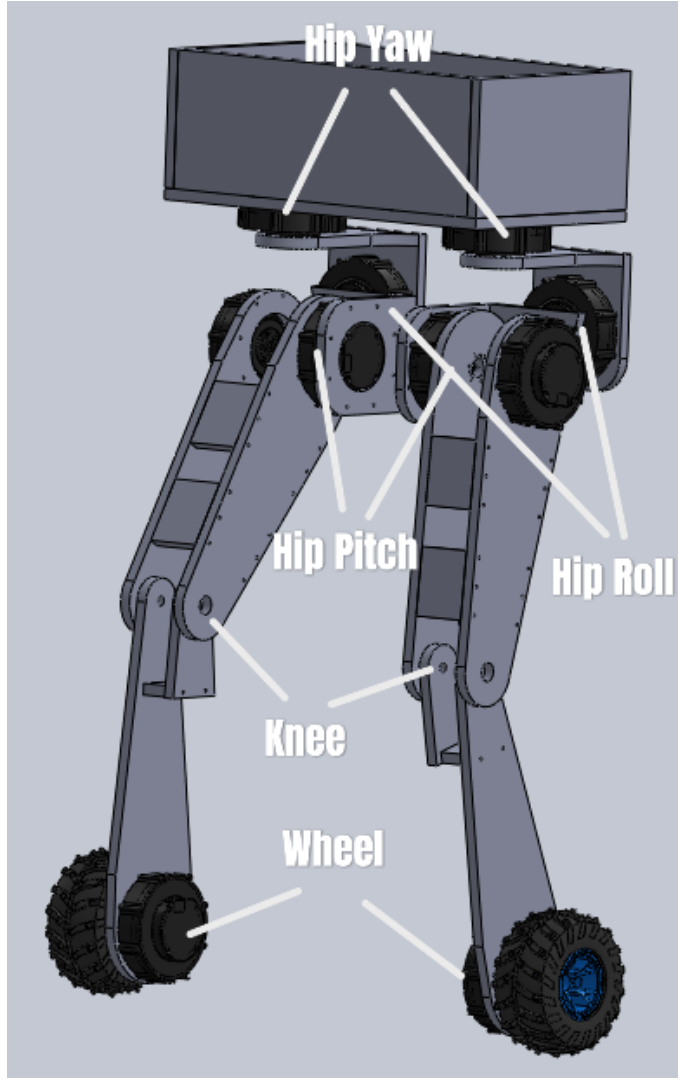

Fig. 1. Mechanical structure and joint layout of the wheeled biped robot. Each leg has five actuated joints: hip roll (HR), hip yaw (HY), hip pitch (HP), knee (KN), and drive wheel (WH). The left leg is annotated; the right leg is symmetric. Figure source: `paper/figures/annotated_robot_joints.png`.

Let the joint vector be defined as

$$q_j = \begin{bmatrix} q^L_{\mathrm{hr}} & q^L_{\mathrm{hy}} & q^L_{\mathrm{hp}} & q^L_{\mathrm{kn}} & q^L_{\mathrm{wh}} & q^R_{\mathrm{hr}} & q^R_{\mathrm{hy}} & q^R_{\mathrm{hp}} & q^R_{\mathrm{kn}} & q^R_{\mathrm{wh}} \end{bmatrix}^\top, \tag{1}$$

where the superscripts L and R denote the left and right legs, respectively.

Fig. 1 illustrates the mechanical structure and joint layout. The articulated hip and knee joints enable variable-height posture regulation, while the wheel joints provide balancing capability through active torque generation.

B. Simulation Environment

All policies are trained in MuJoCo MJX [2], [3] with JAX-based [4] vectorized simulation. The physics simulation runs at $500\,\mathrm{Hz}$ with timestep

$$\Delta t_{\mathrm{sim}} = 0.002\,\mathrm{s}, \tag{2}$$

while the policy is queried at $50\,\mathrm{Hz}$,

$$\Delta t_\pi = 0.02\,\mathrm{s}. \tag{3}$$

Each policy action is therefore held constant for ten simulation substeps. Training is parallelized across 4096 environments on GPU.

Domain randomization [8] is applied during training to improve robustness: body masses are perturbed by $\pm 15\%$, ground friction coefficients by $\pm 40\%$, and joint damping coefficients by $\pm 30\%$ relative to their nominal values. Randomization is applied per episode at reset time. In the base `balance` stage, no external push disturbances are applied ($F_{\mathrm{push}} = 0$); random joint-position noise of $\pm 0.10\,\mathrm{rad}$ is added at reset to expose the policy to varied initial postures. Push disturbances ($40\,\mathrm{N}$ applied over 8 control steps, $\approx 0.16\,\mathrm{s}$) are introduced only in the `balance_robust` fine-tuning stage.

C. Low-Level PID Control

Rather than mapping policy outputs directly to actuator torques, we employ a PID low-level controller that converts normalized policy targets into torque commands. This decouples the policy from raw torque generation and simplifies the learning problem.

The policy outputs $a_t \in [-1, 1]^{10}$, which are interpreted differently depending on the joint type:

- **Leg joints** (hip roll, hip yaw, hip pitch, knee): the policy output is interpreted as a position target. A PD+I controller then computes the torque:

$$\tau_j = K_p^j(q_j^* - q_j) + K_d^j(0 - \dot{q}_j) + K_i^j \int (q_j^* - q_j)\, dt, \tag{4}$$

  where q_j^* is the position target, q_j and $\dot{q}_j$ are the measured joint position and velocity, and K_p^j, K_d^j, K_i^j are per-joint gains. The integral term includes anti-windup clamping ($\pm 0.35\,\mathrm{rad{\cdot}s}$).
- **Wheel joints**: the policy output is interpreted as a velocity target. A PI controller computes the torque:

$$\tau_w = K_p^w(\dot{q}_w^* - \dot{q}_w) + K_i^w \int (\dot{q}_w^* - \dot{q}_w)\, dt, \tag{5}$$

  where $\dot{q}_w^*$ is the velocity target. The derivative gain is masked to zero ($K_d^w = 0$) because wheel velocity-error derivative is the joint acceleration rather than joint velocity, making standard K_d semantics ill-defined for velocity-control joints.

An action-smoothing filter with coefficient $\alpha = 0.4$ blends the current and previous policy output before passing to the PID controller, reducing command chattering. Optionally, an action delay of n_{delay} control steps can model CAN-bus latency during sim-to-real preparation (default: $n_{\mathrm{delay}} = 0$ for simulation training). All torque outputs are clipped to the actuator limits. The per-joint gains are given in Table I.

TABLE I
PER-JOINT PID GAINS. JOINT ORDER: HIP ROLL (HR), HIP YAW (HY), HIP PITCH (HP), KNEE (KN), WHEEL (WH). GAINS ARE IDENTICAL FOR LEFT AND RIGHT LEGS. THE WHEEL K_d IS MASKED TO ZERO INTERNALLY REGARDLESS OF THE TABLE VALUE.

Joint	K_p	K_i	K_d
HR	55.0	0.8	3.0
HY	40.0	0.4	2.0
HP	70.0	1.0	4.0
KN	70.0	1.0	4.0
WH	4.0	0.1	(0.0)

IV. PROBLEM FORMULATION

A. State, Observation, and Action

At time step t, the robot state is $x_t = \{q_t, \dot{q}_t\}$, where q_t and $\dot{q}_t$ are the generalized coordinates and velocities.

The policy receives a **41-dimensional** observation vector

$$o_t = \left[g_t^b,\ v_t^b,\ \omega_t^b,\ q_{j,t},\ \dot{q}_{j,t},\ a_{t-1},\ \bar{h}_t^{cmd},\ e_{\psi,t}\right] \in \mathbb{R}^{41}, \tag{6}$$

where:

- $g_t^b \in \mathbb{R}^3$: gravity vector expressed in the body frame (from IMU accelerometer);
- $v_t^b \in \mathbb{R}^3$: body-frame linear velocity;
- $\omega_t^b \in \mathbb{R}^3$: body-frame angular velocity;
- $q_{j,t} \in \mathbb{R}^{10}$: joint positions;
- $\dot{q}_{j,t} \in \mathbb{R}^{10}$: joint velocities;
- $a_{t-1} \in \mathbb{R}^{10}$: previous smoothed policy action;
- $\bar{h}_t^{cmd} \in \mathbb{R}$: normalized height command;
- $e_{\psi,t} \in \mathbb{R}$: yaw error (heading drift from episode start).

The base 39-dimensional observation $[g_t^b,\ v_t^b,\ \omega_t^b,\ q_{j,t},\ \dot{q}_{j,t},\ a_{t-1}]$ is inherited from the base environment. The two task-specific extensions are:

Normalized height command.

$$\bar{h}_t^{cmd} = \frac{h_t^{cmd} - h_{\min}}{h_{\max} - h_{\min}}, \tag{7}$$

where $h_{\min} = 0.40\,\mathrm{m}$ and $h_{\max} = 0.70\,\mathrm{m}$. The normalization always uses the full task range, so the value is in $[0, 1]$ regardless of the current curriculum level.

Yaw error.

$$e_{\psi,t} = \mathrm{wrap}(\psi_t - \psi_0)\,, \tag{8}$$

where ψ_t is the current torso yaw extracted from the base quaternion via Euler decomposition, ψ_0 is the yaw at episode reset, and $\mathrm{wrap}(\cdot)$ maps to $(-\pi, \pi]$. This term is necessary because the gravity-body-frame component g_t^b is yaw-invariant: a memoryless MLP cannot perceive accumulated heading drift from g^b and ω^b alone.

When `lin_vel_mode="disabled"` is configured for sim-to-real preparation without a reliable velocity estimator, v_t^b is removed, reducing the observation to 38 dimensions. All results in this paper assume the default `lin_vel_mode="clean"`.

Observations are normalized online using Welford's running mean and variance estimator [7], updated during training rollouts and held fixed during evaluation.

TABLE II
REWARD COMPONENTS AND WEIGHTS FOR THE VARIABLE-HEIGHT BALANCE TASK. WEIGHTS FOR THE BALANCE_ROBUST FINE-TUNING STAGE DIFFER: $w_{\text{still}} = 0$, $w_{\text{wh}} = 0$, $w_{\text{nat}} = 1.5$, $w_{\text{alive}} = 0.5$, $w_{\Delta a} = -0.005$, TO ACCOMMODATE WHEEL-BASED PUSH RECOVERY MOTION.

Component	Type	Weight
Height tracking (r_{height})	Positive	2.5
Body level (r_{level})	Positive	1.5
Position drift (r_{drift})	Positive	1.5
Orientation (r_{orient})	Positive	0.8
Symmetry (r_{sym})	Positive	1.0
Heading (r_{heading})	Positive	0.5
Legs forward (r_{fwd})	Positive	0.5
Legs vertical (r_{vert})	Positive	0.5
No motion (r_{still})	Positive	0.5
Yaw suppression (r_{yaw})	Positive	0.5
Natural pose (r_{nat})	Positive	0.4
Alive bonus (r_{alive})	Positive	0.3
Action rate ($p_{\Delta a}$)	Penalty	-0.05
Wheel velocity (p_{wh})	Penalty	-0.005
Joint torque (p_{τ})	Penalty	-0.0005
Joint velocity ($p_{\dot{q}}$)	Penalty	-0.00061

The policy outputs a normalized action $a_t = \pi_\theta(o_t) \in [-1, 1]^{10}$, converted to actuator torques through the PID controller described in Section III-C.

B. Learning Objective

We seek a policy π_θ that maximizes the expected discounted return

$$J(\theta) = \mathbb{E}_{\pi_\theta}\left[\sum_{t=0}^{T-1} \gamma^t r_t\right], \tag{9}$$

where $\gamma = 0.99$ and r_t is the reward at step t.

Training is performed using PPO [1] with clipped surrogate objective

$$L^{\text{PPO}}(\theta) = \mathbb{E}_t\Big[\min\Big(\rho_t(\theta)\hat{A}_t,\ \text{clip}\big(\rho_t(\theta), 1-\epsilon, 1+\epsilon\big)\hat{A}_t\Big)\Big], \tag{10}$$

where $\rho_t(\theta) = \pi_\theta(a_t|o_t)/\pi_{\theta_{\text{old}}}(a_t|o_t)$ and $\epsilon = 0.2$. The full objective adds a clipped value loss (coefficient 0.5) and entropy regularization (coefficient 0.005). Advantages $\hat{A}_t$ are computed using Generalized Advantage Estimation (GAE) [6] with $\lambda = 0.95$.

V. REWARD DESIGN

A. Reward Structure

The total reward at each step is a weighted sum:

$$r_t = \sum_i w_i\, r_t^{(i)}. \tag{11}$$

The complete set of reward terms and weights for the `balance` task are given in Table II.

B. Primary Reward Terms

1) Body-Level Reward: To encourage the torso to remain parallel to the ground:

$$r_{\text{level}} = \exp\left(-\frac{\phi_t^2}{\sigma_r^2}\right)\exp\left(-\frac{\theta_t^2}{\sigma_p^2}\right), \tag{12}$$

where ϕ_t and θ_t are the torso roll and pitch angles, and $\sigma_r = \sigma_p = 0.15\,\text{rad}$.

2) Height Tracking Reward:

$$r_{\text{height}} = \exp\left(-\frac{(h_t - h_t^{cmd})^2}{\sigma_h^2}\right), \tag{13}$$

with $\sigma_h = 0.25\,\text{m}$, providing a gradient signal even when the actual height is far from the target.

3) Heading and Position Regularization: To keep the robot near its initial yaw ψ_0 and anchor position $p_{xy,0}$:

$$r_{\text{heading}} = \exp\left(-\frac{\text{wrap}(\psi_t - \psi_0)^2}{\sigma_\psi^2}\right), \tag{14}$$

$$r_{\text{drift}} = \exp\left(-\frac{\|p_{xy,t} - p_{xy,0}\|_2^2}{\sigma_d^2}\right), \tag{15}$$

with $\sigma_\psi = 0.5\,\text{rad}$ and $\sigma_d = 0.5\,\text{m}$.

4) Orientation and Yaw Stability: Linear orientation stability (avoids saturation at high angular rates):

$$r_{\text{orient}} = \text{clip}\big(1 - 0.15\,\|\omega_t^b\|_2,\ 0,\ 1\big), \tag{16}$$

and a dedicated yaw suppression term:

$$r_{\text{yaw}} = \text{clip}\big(1 - 0.5\,|\omega_{z,t}^b|,\ 0,\ 1\big). \tag{17}$$

5) Leg Symmetry:

$$r_{\text{sym}} = \exp\left(-\frac{\left\|q_{\text{leg},t}^L - q_{\text{leg},t}^R\right\|_2^2}{\sigma_{\text{sym}}^2}\right), \tag{18}$$

where $q_{\text{leg}}^{L/R}$ includes hip roll, hip yaw, hip pitch, and knee for each leg, and $\sigma_{\text{sym}} = 0.5$.

6) Leg Alignment Rewards: Two terms encourage well-aligned leg orientation:

Legs forward (r_{fwd}): rewards hip yaw close to zero (feet pointing forward):

$$r_{\text{fwd}} = \exp\left(-\frac{(q_{\text{hy}}^L)^2}{\sigma_\ell^2}\right)\exp\left(-\frac{(q_{\text{hy}}^R)^2}{\sigma_\ell^2}\right). \tag{19}$$

Legs vertical (r_{vert}): rewards hip roll close to zero (legs perpendicular to the ground plane):

$$r_{\text{vert}} = \exp\left(-\frac{(q_{\text{hr}}^L)^2}{\sigma_\ell^2}\right)\exp\left(-\frac{(q_{\text{hr}}^R)^2}{\sigma_\ell^2}\right), \tag{20}$$

where $\sigma_\ell = 0.15\,\text{rad}$ for both terms.

7) No-Motion Reward: To encourage stationary balance:

$$r_{\text{still}} = \exp\left(-\frac{\|v_t^b\|_2^2}{\sigma_v^2}\right), \tag{21}$$

with $\sigma_v = 0.2\,\text{m/s}$. This term is set to zero in the `balance_robust` stage, where wheel motion is needed for push recovery.

C. Height-Consistent Natural Pose Reward

A key design of our method is a posture regularization term that enforces consistency between the commanded torso height and the desired leg configuration. Let the desired hip-pitch and knee angles $q^*_{\text{hp}}(h^{cmd})$ and $q^*_{\text{kn}}(h^{cmd})$ be computed by piecewise-linear interpolation.

The kinematic keypoints, derived from forward kinematics of the two-link leg model, are:

$$\begin{aligned} h = 0.72\,\text{m}: &\quad q^*_{\text{hp}} = 0.0, \quad q^*_{\text{kn}} = 0.0 \\ h = 0.71\,\text{m}: &\quad q^*_{\text{hp}} = 0.3, \quad q^*_{\text{kn}} = 0.5 \\ h = 0.38\,\text{m}: &\quad q^*_{\text{hp}} = 1.5, \quad q^*_{\text{kn}} = 2.5 \end{aligned}$$

For the high-standing segment $h^{cmd} \in [0.71, 0.72]\,\text{m}$:

$$t_h = \frac{h^{cmd} - 0.71}{0.01}, \quad q^*_{\text{hp}} = 0.3(1 - t_h), \quad q^*_{\text{kn}} = 0.5(1 - t_h). \tag{22}$$

For the lower-height segment $h^{cmd} \in [0.38, 0.71]\,\text{m}$ (which covers the full `balance` training range):

$$t_l = \frac{h^{cmd} - 0.38}{0.33}, \quad q^*_{\text{hp}} = 1.5 - 1.2\,t_l, \quad q^*_{\text{kn}} = 2.5 - 2.0\,t_l. \tag{23}$$

The posture error is

$$e_{\text{pose}} = \Big((q^L_{\text{hp}} - q^*_{\text{hp}})^2 + (q^R_{\text{hp}} - q^*_{\text{hp}})^2 + (q^L_{\text{kn}} - q^*_{\text{kn}})^2 + (q^R_{\text{kn}} - q^*_{\text{kn}})^2\Big)^{1/2} \tag{24}$$

The natural pose reward is

$$r_{\text{nat}} = \exp\left(-\frac{e^2_{\text{pose}}}{\sigma^2_{nat}}\right) r_{\text{knee}}, \tag{25}$$

where $\sigma_{nat} = 0.8\,\text{rad}$ and r_{knee} penalizes knee hyperextension:

$$r_{\text{knee}} = \exp\left(-\frac{\min(q^L_{\text{kn}}, 0)^2}{\sigma^2_k}\right) \exp\left(-\frac{\min(q^R_{\text{kn}}, 0)^2}{\sigma^2_k}\right), \tag{26}$$

with $\sigma_k = 0.15\,\text{rad}$.

This reward prevents the policy from exploiting unnatural joint configurations (e.g., extreme hip abduction with nearly straight knees) to satisfy a height target without a kinematically meaningful posture.

D. Effort and Smoothness Penalties

We regularize torque magnitude, joint velocity, action variation, and wheel rotation:

$$p_\tau = \sum_{j=1}^{10} \tau^2_{t,j}, \quad p_{\dot{q}} = \sum_{j=1}^{10} \dot{q}^2_{j,t}, \quad p_{\Delta a} = \sum_{j=1}^{10} (a_{t,j} - a_{t-1,j})^2, \tag{27}$$

$$p_{\text{wh}} = \dot{q}^{L\,2}_{\text{wh},t} + \dot{q}^{R\,2}_{\text{wh},t}. \tag{28}$$

The wheel velocity penalty and r_{still} together enforce stationary balance in the `balance` stage; both are set to zero in `balance_robust`.

E. Termination Conditions

An episode terminates early if the torso tilt exceeds $0.8\,\text{rad}$ ($\approx 46°$) or the torso height drops below $0.3\,\text{m}$. Otherwise, episodes run for 1000 steps ($\approx 20\,\text{s}$). A survival bonus $r_{\text{alive}} \in \{0, 1\}$ is given at each non-terminal step, providing a constant per-step incentive to remain standing.

VI. Curriculum Learning Strategy

A. Within-Stage Height Curriculum

Rather than sampling the commanded height from the full feasible range from the beginning, we employ a within-stage curriculum [9] that progressively expands the lower bound of the height command distribution.

The height lower bound is decreased by a fixed step δ_h whenever performance exceeds a threshold:

$$h^{(k+1)}_{\min} = \max\big(h^{(k)}_{\min} - \delta_h,\ h^{\text{final}}_{\min}\big), \quad \text{if } \bar{r}_{\text{eval}} \geq r_{\text{thr}}. \tag{29}$$

Advancement signal. Every $I_{\text{eval}} = 2$ training updates, the current policy is evaluated greedily (no exploration noise) over 200 complete episodes using 32 parallel environments, with the observation normalizer held fixed. The per-step return is

$$\bar{r}_{\text{eval}} = \frac{R_{\text{mean}}}{T}, \tag{30}$$

where R_{mean} is the mean episode return and $T = 1000$ is the episode length. Advancement fires when $\bar{r}_{\text{eval}} \geq r_{\text{thr}} = 0.75$. Using a greedy evaluation pass rather than the noisy per-update training reward produces a more reliable signal for curriculum decisions.

Curriculum parameters. Initial lower bound: $h^{\text{init}}_{\min} = 0.69\,\text{m}$; final lower bound: $h^{\text{final}}_{\min} = 0.40\,\text{m}$; $N_c = 29$ discrete levels; $\delta_h \approx 0.01\,\text{m}$; threshold $r_{\text{thr}} = 0.75$. At each level k:

$$h^{cmd}_t \sim \mathcal{U}\big(h^{(k)}_{\min},\ h_{\max}\big), \tag{31}$$

where $h_{\max} = 0.70\,\text{m}$.

The 29 levels span three qualitative phases:

- **Phase A** (levels 1–5): $h_{\min} \in [0.65, 0.69]\,\text{m}$; narrow band near nominal; policy focuses on stable upright stance.
- **Phase B** (levels 6–20): $h_{\min} \in [0.50, 0.64]\,\text{m}$; widening toward moderate crouching.
- **Phase C** (levels 21–29): $h_{\min} \in [0.40, 0.49]\,\text{m}$; full operating range including deep crouching.

B. Multi-Stage Macro-Curriculum

The codebase includes a multi-stage curriculum manager (`wheeled_biped/training/curriculum.py`) that chains training stages with warm-starting and performance-gated promotion/demotion. The three stages are:

1) **`balance`** (this paper): variable-height stationary balance with within-stage height curriculum as described above. No external push disturbances ($F_{\text{push}} = 0$). Macro-curriculum promotion threshold: 80% success rate with $w_{\text{eval}} = 7.0\,\text{reward/step}$.
2) **`stand_up`**: height transitions and recovery from non-nominal initial poses. Initialized (warm-started) from

the `balance` checkpoint. Promotion threshold: 80% success rate with $w_{\text{eval}} = 5.0\,\text{reward/step}$.

3) **`balance_robust`**: fine-tuning under periodic random push disturbances ($F_{\text{push}} = 40\,\text{N}$, applied over 8 control steps, every 200 steps $\approx 4\,\text{s}$). Initialized from the `stand_up` checkpoint. Reward weights modified: $w_{\text{still}} = w_{\text{wh}} = 0$; w_{nat} increased to 1.5; $w_{\Delta a}$ reduced to -0.005. Promotion threshold: 80% success rate with $w_{\text{eval}} = 6.0\,\text{reward/step}$.

Demotion fires when success rate falls below 30% within a window of 100 evaluation episodes. Stages `stand_up` and `balance_robust` have corresponding environment implementations and training configurations but have not yet been trained; their results are not reported here.

VII. Classical LQR Baseline

A. Motivation

Classical control methods provide an interpretable reference for evaluating RL policies. We implement an LQR baseline (`wheeled_biped/controllers/lqr_balance.py`) that operates on the same 41-dimensional observation interface and shares the same PID low-level controller as the RL policy, enabling a controlled comparison without confounding actuator-interface effects.

B. Linearized TWIP Model

The sagittal balance component is based on a linearized two-wheeled inverted pendulum (TWIP) model. The state and input are

$$x_{\text{lqr}} = \begin{bmatrix} \theta_{\text{pitch}} \\ \dot{\theta}_{\text{pitch}} \\ v_{\text{fwd}} \\ p_{\text{fwd}} \end{bmatrix}, \quad u = \bar{\omega}_{\text{wheel}}, \tag{32}$$

where θ_{pitch} is the forward lean angle, v_{fwd} is the forward body velocity, p_{fwd} is the forward position drift, and u is the average wheel velocity command.

The continuous-time linearization $\dot{x}_{\text{lqr}} = A\,x_{\text{lqr}} + B\,u$ uses robot constants: mass $m = 8.1\,\text{kg}$, nominal CoM height $l_c = 0.54\,\text{m}$ (updated per height command via IK), and wheel radius $r_w = 0.06\,\text{m}$.

The LQR gain is solved from the continuous algebraic Riccati equation (CARE) with state cost $Q = \text{diag}(10, 2, 3, 0.3)$ and input cost $R = 0.8$:

$$K_{\text{lqr}} = R^{-1}B^\top P, \quad A^\top P + PA - PBR^{-1}B^\top P + Q = 0. \tag{33}$$

C. Height-Dependent Inverse Kinematics

To extend the LQR to variable commanded heights, we perform an offline IK scan over hip-pitch angles from 0 to $1.35\,\text{rad}$, with the knee set to $q_{\text{kn}} = 2\,q_{\text{hp}}$ (symmetric two-link fold geometry). The resulting height–angle mapping is fit by a degree-2 polynomial, yielding target joint angles for any commanded height in $[0.40, 0.70]\,\text{m}$. The effective CoM height $l_c(h^{cmd})$ fed into the TWIP linearization is updated accordingly.

D. Multi-Axis Control

Lateral and yaw axes are controlled by independent PD terms:

$$u_{\text{lat}} = K_{p,\text{roll}} \cdot e_{\text{roll}} + K_{d,\text{roll}} \cdot \dot{e}_{\text{roll}}, \tag{34}$$

$$u_{\text{yaw}} = K_{p,\text{yaw}} \cdot e_{\psi} + K_{d,\text{yaw}} \cdot \omega_z, \tag{35}$$

with $K_{p,\text{roll}} = 0.4$, $K_{d,\text{roll}} = 0.08$, $K_{p,\text{yaw}} = 2.5$, $K_{d,\text{yaw}} = 0.2$. The final wheel command is the superposition of sagittal LQR, lateral PD, and yaw PD outputs, clipped to $\pm 20.0\,\text{rad/s}$.

E. Assumptions and Limitations

The LQR baseline has the following known limitations:

1) **Linear regime only.** The TWIP linearization is valid near the nominal standing height and small lean angles ($|\theta_{\text{pitch}}| \lesssim 10°$). At large lean deviations or very low commanded heights, accuracy degrades.
2) **No lean integral.** Without an integral term on lean error, constant disturbances (friction variation, slopes) produce nonzero steady-state lean.
3) **No large-disturbance recovery.** Under strong pushes the state leaves the linear regime; the LQR provides no recovery mechanism.
4) **Decoupled axes.** Sagittal, lateral, and yaw dynamics are treated independently, ignoring cross-coupling during aggressive maneuvers.

The LQR baseline is therefore a fair reference for the `nominal` and `narrow_height` scenarios but is expected to degrade at full height range, strong pushes, and low friction.

VIII. Training Pipeline and Evaluation Protocol

A. Network Architecture

Separate MLPs for policy and value function: hidden sizes $\{256, 256, 128\}$, ELU activation, orthogonal weight initialization, implemented in Flax [5]. The policy outputs the mean of a Gaussian action distribution with a learned log-standard-deviation initialized to $\log(0.2)$.

B. Training Configuration

PPO with 4096 parallel environments, rollout length 128 steps, 4 epochs per update, 32 minibatches, learning rate 2×10^{-4} (Adam), gradient norm clipping 0.5, entropy coefficient 0.005, value-loss coefficient 0.5.

C. Research Evaluation Protocol

Research evaluation is performed offline using a single-environment CPU MuJoCo rollout with per-step telemetry recording (`scripts/eval_balance.py`). This is distinct from the curriculum advancement signal (`PPOTrainer.eval_pass()`) used during training.

Evaluation scenarios. Ten scenario groups are defined:

1) **Nominal**: flat floor, $h^{cmd} = 0.65\,\text{m}$, standard domain randomization.
2) **Narrow height**: $h^{cmd} = 0.69\,\text{m}$ (Phase A, near-nominal).

3) **Wide height**: $h^{cmd} = 0.60\,\mathrm{m}$ (Phase B, moderate crouching).
4) **Full range**: $h^{cmd} \sim \mathcal{U}[0.40, 0.70]$ (Phase C, full range).
5) **Push recovery**: single $50\,\mathrm{N}$ impulse at $t = 3\,\mathrm{s}$; measures recovery time and maximum recoverable disturbance via binary search (range $10\,\mathrm{N}$ to $300\,\mathrm{N}$, 8 iterations, 50% survival threshold).
6) **Friction low**: friction multiplier $0.6\times$.
7) **Friction high**: friction multiplier $1.4\times$.
8) **Sensor noise + delay**: IMU/encoder noise plus 1-step ($\approx 20\,\mathrm{ms}$) action delay.
9) **Push sweep**: parametric sweep over $\{20, 40, 60, 80, 100, 130, 160, 200\}\,\mathrm{N}$ for degradation curves.
10) **Friction sweep**: parametric sweep over friction multipliers $\{0.3, 0.5, 0.7, 1.0, 1.3, 1.8\}$.

Evaluation metrics. Ten metrics are computed per scenario:

1) Survival time [s]
2) Pitch RMS [deg]
3) Roll RMS [deg]
4) Pitch-rate RMS [rad/s]
5) XY drift max [m]
6) Height RMSE [m]
7) Wheel speed RMS [rad/s]
8) Torque RMS [N·m]
9) Recovery time [s] (push_recovery only; NaN otherwise)
10) Max recoverable push [N] (push_recovery only)

Aggregate metrics: fall rate, survival rate.

Multi-seed evaluation protocol. Final RL results are reported as mean ± standard deviation over three independent training runs with seeds 42, 113, and 999. Each seed produces a fully independent run: the policy is initialized with a distinct random seed and trained from scratch; seeds are never mixed during training or evaluation. The three trained checkpoints reside at:

- `outputs/balance/rl/seed42/checkpoints/final/`
- `outputs/balance/rl/seed113/checkpoints/final/`
- `outputs/balance/rl/seed999/checkpoints/final/`

Aggregated evaluation results (all three seeds) are collected by `scripts/eval_balance.py` with `--checkpoint` pointing to all three directories simultaneously, writing aggregated outputs to `outputs/balance/rl/paper_eval/`. The LQR baseline (evaluation-only; no training) writes to `outputs/balance/lqr/`.

IX. Experimental Results

Training has not yet been completed. This section will be populated with quantitative results after the `balance` policy is fully trained and evaluated. All placeholders below indicate planned experiments.

A. Learning Curves

[TODO-FIGURE: training curve and curriculum level progression]

TABLE III
RL policy vs. LQR baseline comparison (planned). All result cells to be filled after training.

	Scenario	Surv. (%)	e_h (m)	Pitch (deg)	Torque (Nm)	Push (N)
RL	Nominal	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
	Narrow	[TODO]	[TODO]	[TODO]	[TODO]	—
	Full range	[TODO]	[TODO]	[TODO]	[TODO]	—
	Push recov.	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
	Friction low	[TODO]	[TODO]	[TODO]	[TODO]	—
LQR	Nominal	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
	Narrow	[TODO]	[TODO]	[TODO]	[TODO]	—
	Full range	[TODO]	[TODO]	[TODO]	[TODO]	—
	Push recov.	[TODO]	[TODO]	[TODO]	[TODO]	[TODO]
	Friction low	[TODO]	[TODO]	[TODO]	[TODO]	—

TABLE IV
Variable-height balancing: performance vs. commanded height. All values pending training completion.

h^{cmd} (m)	Surv. (%)	e_h (m)	Pitch (deg)	Torque (Nm)
0.40	[TODO]	[TODO]	[TODO]	[TODO]
0.50	[TODO]	[TODO]	[TODO]	[TODO]
0.60	[TODO]	[TODO]	[TODO]	[TODO]
0.65	[TODO]	[TODO]	[TODO]	[TODO]
0.69	[TODO]	[TODO]	[TODO]	[TODO]

B. Nominal Balance and Variable-Height Results

C. Push Robustness

[TODO-FIGURE: push_sweep degradation curve; TODO-RESULT: max recoverable push force]

D. Friction Robustness

[TODO-FIGURE: friction_sweep degradation curve]

E. Ablation: Natural Pose Reward

[TODO-ABLATION: natural pose reward (r_{nat}) ablation: weight=0.4 vs. weight=0]

F. Ablation: Observation Design

[TODO-ABLATION: lin_vel_mode comparison (clean/noisy/disabled) — only report if experiment is run]

X. Discussion and Limitations

A. Framework Design

The main design principle is to decompose the variable-height balancing problem into a progressive curriculum that stabilizes learning by starting near the nominal standing pose and expanding toward more challenging configurations. The height-consistent natural pose reward constrains the joint-space solution to physically meaningful postures at each target height.

The PID low-level controller decouples the RL policy from actuator-level dynamics and is shared with the LQR baseline, ensuring that observed performance differences reflect controller intelligence rather than interface differences.

The yaw-error observation term $e_{\psi,t}$ closes the heading loop that gravity-body-frame observations alone cannot. Without it, a memoryless MLP requires undesirable implicit integration of angular velocity over many steps to sense accumulated yaw drift.

B. Current Limitations

1) **Balance only (experimentally).** The experimentally mature component is the `balance` stage. Stages `stand_up` and `balance_robust` have code and configurations but no training results at the time of writing. All claims about push recovery and height transitions are structural, not performance-validated.
2) **No quantitative results yet.** Training runs have not been completed. All result sections contain placeholders. Claims in this paper concern framework design, not demonstrated performance.
3) **No sim-to-real transfer.** All work is conducted in simulation. Although the robot model matches a physical platform, sim-to-real transfer has not been attempted. The sensor-noise and action-delay configurations are designed for sim-to-real preparation but have not been hardware-validated.
4) **Multi-seed training planned, not yet complete.** The evaluation protocol uses 3 independent seeds (42, 113, 999), each producing a separate training run stored under `outputs/balance/rl/seed{42,113,999}/`. Quantitative results will report mean $\pm$ std across all three seeds.
5) **LQR at large disturbances.** The LQR is only valid near the nominal height and small lean angles. Comparisons at full range, strong pushes, and low friction are expected to favor RL, but this must be confirmed experimentally.
6) **Curriculum advancement signal noise.** The within-stage height curriculum fires every 2 training updates using 32 envs $\times$ 200 episodes. Early-training evaluations may produce noisy advancement signals before the policy stabilizes.

XI. Conclusion

This paper presented a curriculum reinforcement learning framework for variable-height balancing of a 10-actuated wheeled biped robot. The framework includes: (1) a 41-dimensional observation with explicit yaw-error feedback; (2) a PID low-level controller shared with a classical LQR baseline; (3) a within-stage height curriculum gated by a greedy evaluation pass every 2 training updates, with $r_{\mathrm{thr}} = 0.75$; (4) a height-consistent natural pose reward via piecewise-linear kinematic interpolation; and (5) a multi-stage macro-curriculum with stages `balance`, `stand_up`, and `balance_robust`.

A classical LQR baseline operating on the same 41-dimensional observation interface provides an interpretable reference for near-nominal operation. A ten-scenario evaluation benchmark with ten per-scenario metrics provides the infrastructure for rigorous quantitative comparison.

All infrastructure has been fully implemented in MuJoCo MJX with JAX. At the current stage, the main contribution is the system design, the height-consistent posture reward, and the LQR baseline. Quantitative results will be reported in a future revision.

Future work: (1) complete training and report variable-height balance results with ablations over r_{nat} and `lin_vel_mode`; (2) train and evaluate `stand_up` and `balance_robust`; (3) multi-seed statistical validation; (4) quantitative RL vs. LQR comparison across all ten scenarios; and (5) sim-to-real transfer to the physical robot platform.

Acknowledgment

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